PROTECTIVE COATING FOR CONCRETE

MES-ST-SPC-0008

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Protective Coating Systems for Concrete & Concrete Masonry

1. INTRODUCTION

1.1 Scope

This document specifies the minimum requirements for concrete protective coating for Ma'aden projects.

This document is intended to be used in conjunction with the applicable documents listed herein.

1.2 Purpose

The purpose of this specification is to provide the minimum requirements for materials, surface preparation, application and quality control for protective coatings of concrete.

1.3 Conflicts and Deviations

Any deviation from the requirements stipulated in this document shall go through a waiver approval process as per Ma'aden Capital Investment System (MCIS) guidelines and shall be addressed to Ma'aden's Project Development & Engineering (PDE) team for their review and approval.

2. KEY TERMS AND DEFINITIONS

Employer	Saudi Arabian Mining Company (Ma'aden) and its authorized representatives and respective affiliates
Contractor	Party responsible for supplying materials, plant, labor and supervisory services
Manufacturer	Party who produces and warrants performance of products, materials, and/or items provided in accordance with contract documents
Supplier	Party responsible for supplying concrete materials in accordance with contract documents
SBC	Saudi Building Code
AASHTO	American Association of State Highway and Transportation Officials
ACI	American Concrete Institute
ASTM	American Society for Testing and Materials
BS	British Standards
DFT	Dry Film Thickness

ICRI	International Concrete Repair Institute
SD	Equivalent Air Layer Resistance
SSPC	Steel Structures Painting Council

3. REFERENCED DOCUMENTS

This specification requires compliance with the provisions of the latest revision of the relevant standards and codes of the following authorities and technical organizations, as applicable, and specifications which shall be considered to form part of this specification, where applicable. In general, the Contractor shall be responsible for full compliance with applicable Ma'aden Health, Safety and Environmental Standards (MSHEMs). Any deviation from the MSHEMs shall require written approval from the Employer.

3.1 Statutory Regulations

All work performed and materials furnished shall comply with the applicable statutory regulations, codes and other requirements of the Kingdom of Saudi Arabia or local requirements of Project Site. The Contractor shall ensure that he is fully aware and informed of all Government laws, local ordinances, regulations and all orders and decrees of bodies or tribunals having any jurisdiction or authority over the Project and/or Site, which in any manner affect the Project and those engaged or employed on the Project, or which in any way affect the conduct of the Work. The Contractor shall at all times observe and comply with all such Government and/or Ministry laws, bylaws, ordinances, codes, regulations, orders and decrees.

3.2 Industry Codes and Standards

All equipment and systems shall conform to the following:

3.2.1 *Local Codes and Standards (for Saudi Arabia)*

All Ma'aden facilities shall comply with latest version of all Saudi Arabia applicable local codes and standards including but not limited to:

Saudi Building Code SBC 304 – Structural Concrete Structures

For project locations other than Saudi Arabia, the Contractor shall follow the respective applicable local regulations, codes and standards.

3.2.2 *International Codes and Standards*

3.2.2.1. American Association of State Highway and Transportation Officials (AASHTO)

AASHTO T259	Standard Method of Test for Resistance of Concrete to Chloride Ion Penetration

3.2.2.2. American Society for Testing and Materials

ASTM B117	Standard Practice for Operating Salt Spray (Fog) Apparatus
ASTM C267	Standard Test Methods for Chemical Resistance of Mortars, Grouts and Monolithic Surfacings and Polymer Concretes
ASTM C1202	Standard Test Method for Electrical Indication of Concrete's Ability to Resist Chloride Ion Penetration
ASTM C1305/C1305M	Standard Test Method for Crack Bridging Ability of Liquid-Applied Waterproofing Membrane
ASTM D870	Standard Practice for Testing Water Resistance of Coatings Using Water Immersion
ASTM D4258	Standard Practice for Surface Cleaning Concrete for Coating
ASTM D4259	Standard Practice for Preparation of Concrete by Abrasion Prior to Coating Application
ASTM D4262	Standard Test Method for pH of Chemically Cleaned or Etched Concrete Surfaces
ASTM D4263	Standard Test Method for Indicating Moisture in Concrete by the Plastic Sheet Method
ASTM D4541	Standard Test Method for Pull-off Strength of Coatings Using Portable Adhesion Testers
ASTM D4787	Standard Practice for Continuity Verification of Liquid or Sheet Linings Applied to Concrete Substrates.
ASTM D5589	Standard Test Method for Determining the Resistance of Paint Films and Related Coatings to Algal Defacement

ASTM D6489	Standard Test Method for Determining the Water Absorption of Hardened Concrete Treated With a Water Repellent Coating
ASTM D7234	Standard Test Method for Pull-Off Strength of Coatings on Concrete Using Portable Pull-Off Adhesion Testers
ASTM E96/E96M	Standard Test Methods for Gravimetric Determination of Water Vapor Transmission Rate of Materials
ASTM E337	Standard Test Method for Measuring Humidity with a Psychrometer (the Measurement of Wet- and Dry- Bulb Temperatures)
ASTM G152	Standard Practice for Operating Open Flame Carbon Arc Light Apparatus for Exposure of Nonmetallic Materials
ASTM G153	Standard Practice for Operating Enclosed Carbon Arc Light Apparatus for Exposure of Nonmetallic Materials

3.2.2.3. American Concrete Institute

ACI PRC-224	Control of Cracking in Concrete Structures
ACI PRC-503	Use of Epoxy Compounds with Concrete
ACI PRC-515.1	Guide to the use of Waterproofing, Damp proofing, Protective, and Decorative Barrier Systems for Concrete
ACI PRC-515.2	Guide to Selecting Protective Treatments for Concrete
ACI PRC-515.3	Guide for Assessment and Surface Preparation for Application of Protection Systems for Concrete

3.2.2.4. Process Industry Practices (PIP)

PIP CTSE1000	Application of External Coatings
PIP CTSE1000-D1	Documentation Requirements Sheet
PIP CTCE1000-D2	Coating Type Code Definitions
PIP CTCE1000-D12M	Severe/Seacoast Environment Selection Criteria (SI Units)
PIP CTSE1000-F	Daily Inspection Report Form

3.2.2.5. National Association of Corrosion Engineers (NACE)

NACE SP0188	Discontinuity (Holiday) Testing of New Protective Coatings on Conductive Substrates

3.2.2.6. British Standard

BS EN 12504-2	Testing concrete in structures, Part 2: Non-destructive testing - Determination of rebound number
BS EN 12504-4	Testing concrete in structures, Part 4: Determination of ultrasonic pulse velocity
BS EN 1062-6	Paints and varnishes-Coating Materials and Coating Systems for Exterior Masonry and Concrete, Part 6: Determination of Carbon Dioxide Permeability

3.2.2.7. Steel Structures Painting Council (SSPC)

SSPC-TU 2	Design, Installation, and Maintenance of Coating Systems for Concrete used in Secondary Containment
SSPC-SP 13	Surface Preparation of Concrete

3.2.2.8. International Standardizations Organization (ISO)

ISO 4628-2	Paints and varnishes-Evaluation of degradation of coatings. Designation of quantity and size of defects, and of intensity of uniform changes in appearance Part 2: Assessment of degree of blistering
ISO 4628-4	Paints and varnishes-Evaluation of degradation of coatings-Designation of quantity and size of defects, and of intensity of uniform changes in appearance Part 4: Assessment of degree of cracking
ISO 2409	Paints and varnishes- Cross-cut test

3.2.2.9. Green Seal Standards (GS)

GS-11	Paints, Coatings, Stains, And Sealers

3.2.3 *Alternative Design Codes*

Alternative codes may be used if approved by the Employer.

3.3 Ma'aden Standards

MD-101-SMEM-EG-ST-SPC-0005 Concrete Works

3.4 Site Data

All materials, equipment and work methods shall be suitable for the Site conditions as defined in the applicable Site Data Specification.

4. SUBMITTALS

The Contractor's detailed coating specification including specification of coating Supplier and protective coating system (coatings and joint sealant) shall be submitted to the Employer for approval prior to application. The detailed specification shall be accompanied by an Inspection and Test Plan.

5. MATERIALS

5.1 Coating System

The concrete protective coating system shall be a tough, durable and chemically resistant system which will withstand the nominated operating conditions to give long-term protection of the concrete substrate.

The Protective Coating System shall be designed for a minimum guaranteed design life of:

- 5 years minimum (higher period per project requirements) for the coating system before significant maintenance or replacement, and
- 3 years minimum (higher period per project requirements) for the joint sealing system before significant maintenance or replacement.

The Contractor shall ensure that the bond strength required to prevent delamination of the protective coating system from the concrete substrate is adequate for the specified ambient and operating conditions. In particular, differential temperatures, differences in the coefficients of thermal expansion for the coating system and concrete, and varying shrinkage between the protective coating system and concrete substrate shall be taken into account.

Concrete to which the protective coating system is applied should be designed to ACI PRC-224.3 which should limit the mean crack width to 0.15 mm. The protective coating system shall be capable of and guaranteed to bridge any cracks in the concrete substrate up to 0.3 mm in width.

Floors to receive a protective coating system should, where practicable, have a minimum grade of:

- 1.5% (1 in 67) with an optimal grade of 2% where floors are exposed to solutions.
- 3% if exposed to slurries.

For a list of approved protective coatings for concrete refer Appendix A. Contractor may propose for approval of alternative coating system

5.2 Coating Supplier

Prior to application of any coating material the coating Supplier shall confirm:

- The suitability of its products for its intended use and the proposed installation methods and details for the ambient and operating conditions. In case any temporary cargo need to be stored i.e. sea water during commissioning, the suitability of coating shall be confirmed for temporary cargo also by coating Manufacturer.
- The bond strength of the protective coating system to the concrete substrate is adequate over the range of specified operating conditions to prevent delamination.

Prior to application of any joint sealing material the coating Supplier shall confirm:

- The suitability of its products for its intended use and the proposed installation methods for the ambient and operating conditions
- Compatibility between the joint sealant and the coating system.

5.3 Coating Requirements

The protective coating system for concrete surfaces, including compatible joint sealing systems shall be supplied and installed in accordance with the Contract specification and the material Manufacturer's approved instructions. The coating properties and suitability for different environments and conditions shall comply with the SSPC-TU 2.

The protective coatings shall protect the treated concrete area from degradation due to chemical substances, application of mechanical loads, environmental conditions and both general and specific duty requirements of the areas.

The protective coating system shall have resistance to the chemicals declared for each plant as well as spillage of oil and other petroleum products, paint and associated chemical solutions, cleaning solvents and regular hose-down using hyper-saline plant water.

For each coating system a proposed method for the sealing and protection of concrete isolation joints and saw-cut joints in ground slabs shall be included in the Contractor's detailed coating specification.

The Contractor shall nominate the maximum crack width that may occur in the underlying concrete without reducing the effectiveness of the coating system, i.e. it is expected that a degree of flexibility in the coating will enable it to remain intact despite cracks occurring in the concrete.

The protective coating system shall accommodate the thermal effect arising from the spillage of process slurries and chemicals and environmental conditions, with due consideration given to potential differential thermal expansion between the concrete substrate and the protective coating systems.

The protective coating systems shall be suitable for application to pedestals and sumps subjected to static and dynamic loads and shall have sufficient durability and toughness to sustain the mechanical loads specified.

The Contractor's detailed coating specification shall provide an application process that will ensure a robust sealant/coating interface around sumps, where the most aggressive process conditions will be experienced.

The Contractor to follow requirements mentioned in this specification or coating Manufacturer requirements whichever is more stringent. The method statement prepared by Contractor for execution of coating work shall be approved by Coating Manufacturer as a minimum. Coating shall be applied by applicators approved by Coating Manufacturer. Any exception shall be approved by the Employer.

5.4 Thinner

Thinner, diluents or clean up materials shall be from the same Manufacturer as the specified coatings and shall be the recommended product for that coating.

5.5 Sustainability requirements

5.5.1 *Paints and Coatings*

1. Paints and coatings used on the interior of the building (defined as inside of the waterproofing system and applied on site) shall comply with the following criteria:
   a). Architectural paints, coatings and primers applied to the interior walls and ceilings: Do not exceed the VOC content limits established in the Green Seal Standard GS-11, Paints, Coatings, Stains, and Sealers, Edition 4.0.
      (1) Flats: 50 g/L
      (2) Non-Flats: 150 g/L
      (3) Anti-corrosive and anti-rust paints applied to interior ferrous metal substrates
   b). Clear wood finishes, floor coatings, stains, sealers, and shellacs applied to interior elements: Do not exceed the VOC content limits established in the South Coast Air Quality Management District (SCAQMD) Rule 1113, Architectural Coatings.
      (1) Clear wood finishes:
         (a) Varnish: 275 g/L
         (b) Lacquer 275 g/L
      (2) Floor coatings: 50 g/L
      (3) Sealers:
         (a) Waterproofing sealers: 100 g/L
         (b) Sanding sealers: 275 g/L
         (c) All other sealers: 200 g/L
      (4) Shellacs:
         (a) Clear: 730 g/L
         (b) Pigmented: 550 g/L
      (5) Stains: 250 g/L
2. Aromatic Compounds: Paints and coatings shall not contain more than 1% (by weight) total aromatic compounds (hydrocarbon compounds containing one or more benzene rings).

6. SURFACE PREPARATION

6.1 Examination of Substrate

The Contractor shall examine the concrete surface and shall carry out any testing required to confirm that the substrate is suitable for application of the protective coating system prior to acceptance of the concrete area to be coated. The Contractor shall notify the Employer in writing of any areas that are unacceptable or not suitable for coating.

Concrete to be coated shall generally be at least 28 days old unless the Contractor can demonstrate adequate tensile strength and moisture content of the concrete substrate, and these conditions are acceptable to the coating Supplier.

6.2 Surface Defects

Defects such as formwork fins, sharp edges etc, which may adversely affect the integrity of the Protective Coating System shall be removed by grinding or other mechanical means prior to application of the coating system.

Concrete, particularly vertical surfaces, may contain "Honeycomb" or "blow holes" in or just below the surface which may affect the integrity of the coating system. These defects shall be exposed by abrasive blasting or similar during the surface preparation, and then filled with vinyl ester or epoxy mortar. All surfaces shall be primed prior to filling or other repair.

6.3 Repair of Surface Defects

Cracks less than 1.0 mm wide and holes not exceeding 10 mm in diameter shall be filled with an epoxy filler or grout before applying primer/sealer to the surface. Cracks subject to movement shall be treated as a movement joint, and shall be prepared using methods, materials and procedures recommended by the Manufacturer. All voids shall be prepared, primed and filled using suitable and compatible epoxy filler prior to coating. The epoxy filler materials shall be approved by the Employer and Coating Manufacturer. Surfaces with sharp corners and subject to immersion conditions shall be rounded by forming a fillet with approximately 25 mm radius using an epoxy grout.

6.4 Surface Preparation Methods

Chemical cleaning of the concrete substrate shall be carried out in accordance with ASTM D4258 as required to remove oils and grease and any other contamination from the surface prior to surface

preparation. The surface preparation activity shall adequately remove laitance, contaminants and other adhered compounds on the concrete such as curing agents.

The surface shall be prepared using one or more of the following methods:

- Track blasting
- Captive shot blasting
- Wet abrasive blasting
- Dry abrasive blasting – under special conditions
- Hydro blasting -where pressure is adequate to achieve the nominated surface profile
- Mechanical scabbling
- Diamond wheel grinding.

The method of surface preparation shall be included in the detailed coating specification.

Dry blasting will only be approved under special circumstances where dust can be adequately controlled to the satisfaction of the Employer.

A surface profile (peak-to-valley) amplitude of concrete (excluding aggregate) of 300 µm (approximating G60 abrasive paper) shall be achieved. The surface preparation activity shall include:

Repair of cracks (includes both inactive & active cracks), spalls, pop outs, and other irregularities. Special care to be taken to repair active cracks.

Removal of surface contaminants that inhibit coating adhesion or induce early coating deterioration

Roughening of smooth surfaces to improve coating adhesion

Elimination of any weak surface layers to establish the strong base required.

SSPC-SP 13 Surface Preparation of Concrete shall be used as guidelines. The recommended surface preparation methods to be used for each application type is as noted in Table 6-1.

The Surface preparation method shall also meet Coating Manufacturer recommendations.

Table 6-1 : Recommended Concrete Surface Preparation (This need to be followed in absence of Manufacturer recommendations only)

Application	Profile	Surface Preparation Method
Sealers	0 – 0.076 mm	Detergent Scrub
		Low pressure Water
		Acid Etching (not recommended)
		Grinding
Thin Films	0.102 – 0.254 mm	Acid Etching (not recommended)
		Grinding
		Abrasive Blast
		Steel Shot Blast
High Build	0.254 – 1.016 mm	Abrasive Blast
		Steel Shot Blast
		Scarifying
Self-Leveling	1.27 – 3.128 mm	Abrasive Blast
		Steel Shot Blast
		Scarifying
		Needle Scaling
		High/Ultra high-pressure Water Jetting
Polymer Overlay	3.128 – 16.129 mm	Abrasive Blast
		Steel Shot Blast
		Scarifying
		Needle Scaling
		High/Ultra high-pressure water jetting
		Scabbling
		Flame Blasting
		Milling/rotomilling

6.5 Profile

The profile required for the specific protective coating application system shall be categorized as per ICRI Technical Guideline No 03732, expressed as a Concrete Surface Profile number (CSP 1-9).

The required profile shall be specified according to the project requirements and Coating Manufacturer recommendations.

6.6 Light Duty Service

The concrete surface shall be scoured to remove any protrusions and sharp edges by using an abrasive brick (carborundum stone), power grinding, or power brushing. The overall surface soundness shall be checked by scratching a screwdriver or pocketknife across the surface. If the metal object rides over the surface without loosening any particles and leaves only a shiny mark, the structure is sound. Otherwise, scour shall be continued.

Surface contaminants such as laitance and form oils must be removed. Form oils, grease, etc shall be removed by steam cleaning with a water emulsifying detergent added. Laitance shall be removed by wire brushing. If the concrete surface is quite smooth such as "formed" surfaces, dry and light sweep blasting shall be used to roughen the surface. The surface of the new concrete shall be roughened to the texture of medium grit sandpaper. High pressure (20684 kPa at the nozzle) water rinse injected with abrasive can be used, subject to Employer approval. After cleaning is complete, the concrete surface shall be checked to ensure that surface is free of voids which may be exposed or enlarged during the blasting process. Voids shall be repaired if necessary.

6.7 Heavy Duty Service

The surface shall be prepared as stated for light duty service in Section 6.6. The tensile strength of the concrete surface shall be measured in accordance with ASTM D4541. The "pull-off" shall meet or exceed 1200 kPa for immersion service and 1790 kPa for industrial floors. Higher value of pull-off test shall be followed if recommended by Manufacturer.

6.8 Unacceptable Methods

Chemical cleaning of the concrete to ASTM D4258 may only be carried out to remove oils and grease and other contamination from the surface prior to substrate preparation. Acid washing is not acceptable.

6.9 Protection of Items During Surface Preparation

All electrical cables, wiring, instrumentation, other electrical items and mechanical equipment as necessary shall be protected from the effects of surface preparation processes to the satisfaction of the Employer.

6.10 Approval of Surface for Coating

Surfaces must be clean, dry and sound to the approval of the Employer, prior to application of any primer or coating surface.

Contractor should follow the allowable time interval between surface preparation and application of coating as specified by Coating Manufacturer.

7. APPLICATION OF PROTECTIVE COATINGS AND JOINT SEALANTS

7.1 Atmosphere Conditions

The atmospheric conditions which prevail during the application of coatings and joint sealants shall be such that the surface being coated is completely free of surface moisture.

"Out-gassing" of concrete may occur during periods of increasing temperature gradient which can cause bubbles or holes in the applied concrete repairs or protective coating system. If there is the potential for "out-gassing", it will be necessary to shade the surface from sunlight or apply the material in a decreasing temperature gradient, so the initial cure can take place without "out-gassing" of the concrete occurring.

Coating shall only be applied during favorable weather periods, when rainfall is not expected for the following 12 hours.

The dew point shall be at least 5°C lower than the temperature of the concrete surface before painting commences.

Painting using water-based paints shall not commence whilst the relative humidity is above 85% or where it may be expected to exceed 90% during the 12-hour curing period.

7.2 Unsuitable Conditions

No coatings or joint sealants shall be applied if:

- The ambient temperature is below 5°C, unless otherwise stipulated by the coating supplier's or Joint Sealant Manufacturer's data sheets as acceptable.

- The relative humidity is above 85% or if it is raining
- The substrate temperature is less than 3°C above the dew point
- The substrate temperature is less than 10°C
- The substrate temperature is greater than 28°C unless otherwise stipulated by the coating Supplier or joint sealant Manufacturer as acceptable
- The ambient temperature is above 35°C, unless otherwise stipulated by the coating supplier's or Joint Sealant Manufacturer's data sheets as acceptable
- The ambient conditions or substrate are outside the acceptable ranges nominated on the coating supplier's or joint sealant Manufacturer's data sheets, installation instructions or application instructions as applicable.
- The moisture content in the substrate is more than 4% by volume.
- There is moisture retained behind a 1 m x 1 m polythene sheet, taped securely onto the concrete surface to form a seal for 24 hours.

The ambient conditions shall be recorded before and at the completion of each day's coating (or joint sealing) and at three hourly intervals during coating (or joint sealing) and shall be submitted with daily records as specified in Section 10 of this specification.

7.3 Mixing

All coatings and joint sealant materials shall be mixed, thinned, applied and used strictly in accordance with the coating Manufacturer's or joint sealant Manufacturer's recommendations as applicable.

Coating materials or joint sealant materials which have livered, gelled or otherwise deteriorated shall not be mixed or used.

Mixed materials shall be strained in accordance with the coating Manufacturer's recommendations to remove any undispersed particles. Excessive retention of pigment and other solids on the screen shall be cause for the product to be remixed and re-sieved before use.

7.4 Pot Life of Catalyzed Materials

The pot life of catalyzed materials at the prevailing working temperature shall not be exceeded. When the pot life limit is reached, the spray pot and other containers shall be emptied, remaining material discarded, the equipment and containers cleaned, and new material mixed and catalyzed.

7.5 Coating Film Thickness

The dry film thickness of each coat and the entire system shall be specified.

During coating the wet film thickness shall be regularly checked at least once in every one square meter with an appropriate gauge to ensure that the required wet film thickness is achieved. The results of this check shall be recorded in the relevant quality control documentation.

7.6 Multiple Coats

Where multiple coats of the same type are specified, each successive coat shall show where possible, a distinguishable difference in color to the one over which it is applied.

7.7 Recoating

Recoating times shall be in accordance with the coating Manufacturer's recommendations for the ambient conditions and temperature prevailing at the time of coating.

7.8 Coating Defects

Adhesion of coatings shall be sound throughout. All coatings shall be free of sagging, craters, dry over spray and other defects. If such defects occur, these shall be brought to the attention of the Employer and repaired according to the coating Manufacturer's recommendations and to the approval of the Employer.

Visual cracking, mud cracking, flaking or blistering of the coating systems as defined by ISO 4628-2 and ISO 4628-4, shall not be allowed.

7.9 Marking of Defects

Marking of defective areas shall be made using a marker compatible with the coating over which it is applied. This compatibility shall be confirmed in writing by the coating Supplier.

7.10 Repairs to Protective Coating

The repair of protective coatings (if required) shall be identified as a 'Hold Point' in the Contractor's inspection and test plan. The Contractor shall advise the Employer of the proposed method of repair. Repairs shall not commence until the repair method is approved. All repair methods shall follow Coating Manufactures recommendations.

7.11 Dust-Debris Combination

Coated surfaces contaminated by embedded dust shall be prepared by either sanding or whip blasting and recoated to the specified dry film thickness using the full protective coating system selected.

7.12 Protection of Adjacent Area

Care shall be exercised to protect adjacent areas and equipment from over-spray by masking with paper or plastic sheeting, the erection of screens, barriers, boarding's and drop sheets.

7.13 Surface Not to Be Coated

The following surfaces and materials shall not be blasted, abraded, mechanically damaged or coated unless specified otherwise:

- Stainless steel surfaces
- Electrical cables
- Other surfaces nominated by the Employer.

All materials used to mask areas requiring protection shall be removed upon completion, and the marking and blemishes, if any, shall be cleaned.

8. PROTECTIVE COATING REQUIREMENTS

8.1 Coating Applications Suitability Criteria for Dry Concrete

For existing painted concrete, prior to applying the fresh paint, a test area shall be prepared on the structure to be painted, except where a compatibility test has already been undertaken. Compatibility tests shall be undertaken to establish whether the new paint will bond to the existing paint. The complete paint system shall be applied in accordance with the Manufacturer's instructions, including any primer and undercoats, to an area of not less than 5 m^2.

When the paint system has cured for 14 days, a test of surface adhesion shall be made in accordance with ASTM D4541/ASTM D7234. Scoring (Cut around circumference of dolly) shall be done around the test surface. Extreme care shall be adopted while scoring to prevent micro cracking in the coating, since such cracks may cause reduced adhesion values. Scored samples constitute a different test, and this procedure should be clearly reported with the results. The average bond of 3 dollies to the concrete, via the completed coating system, shall be satisfactory. Suitable non-destructive testing methods shall be used for measurement of the DFT. If the bond strength results

are not satisfied, the concrete surface shall be cleaned using high pressure water jetting, followed by re-application and re-testing of the coating.

8.2 Coating Applications Suitability Criteria for Wet Concrete Surfaces

Paint for application onto wet concrete surface shall be specifically formulated for tolerance to moisture during application, cure and service.

All concrete surfaces to receive paint shall be surface-dry at the time of application, including any cracks or other defects that may hold water. The concrete surface shall leave no mark when a sheet of pale blue blotting paper is pressed uniformly onto the concrete. If concrete surface cannot be surface-dry due to continuously wet conditions, temporary protection shall be provided to encase the concrete and paint applicator and prevent moisture penetration, ensuring adequate ventilation is provided.

Deposits of salt crystals which collect on coated surfaces must be washed off with potable water and the surface allowed to dry, prior to applying further coats of paint.

8.3 Coating Finish Requirements

The coating finish shall comply with the following criteria:

a). The carbon dioxide diffusion resistance of the paint system, measured in accordance with BS EN-1062-6, shall be expressed in terms of an R-value.

b). The DFT used to calculate the R-value for the paint system shall be the minimum DFT to be achieved in service. The target mean DFT shall be greater than the minimum DFT by 33%.

c). The paint shall be applied in minimum two coats. The minimum DFT shall be not less than 15 µm.

d). The SD value of the paint system at the mean DFT shall not exceed 4 m equivalent air layer resistance and shall be measured in accordance with ASTM E96/E96M.

e). The adhesion of the paint system, measured in accordance with ASTM D7234 shall be satisfactory. This requirement shall also be satisfied after 4000 hours artificial weathering in accordance with ASTM G152/G153.

f). The paint system shall not support algal growth, when measured in accordance with ASTM D5589.

g). Water uptake by the paint system shall be zero when measured in accordance with ASTM D6489.

8.4 Back Pressure Acting on Coatings

Backpressure occurs when concrete is saturated on one face, e.g. exposed to the sea spray.

Where this occurs, the coating must not form a barrier to the passage of water vapor through the concrete, otherwise blistering of the coating may occur. The thickest area of the coating will be at greatest risk from blistering; hence the maximum thickness shall be specified for testing in accordance with ASTM E96/E96M.

The maximum DFT shall not be greater than 1.5 times the target mean DFT.

The water permeability shall be measured in accordance with the requirements in ASTM D870.

The coating shall be considered resistant to water permeation if water does not appear through the coated face on either test.

8.5 Concrete Subjected to Chloride and Carbonation

The paint system shall comply with Section 8.3. The coating applied to structures in the splash zone and atmospheric zone should have good resistance to UV light, abrasion, and be suitable for application under high humidity/moisture conditions. Aesthetic considerations are normally of secondary importance.

Suitable coating systems include acrylic and polyurethane. Epoxy and tar epoxy may also be used if protected from sunlight. Coating systems not appropriate for wet applications and/or splash zones are water-based coatings and soft coatings such as bituminous coatings.

Water-repellent systems such as silanes are not suitable for lower grade concrete, as rapid carbonation may occur. However, for new concrete construction of Grade 45 and above or where carbonation is not a concern, the R-value requirement may be waived, allowing the use of water-repellent systems. The non-film forming water repellent system should be specified to comply with the Manufacturer's minimum recommended coverage rate. The treated concrete should be resistant to water uptake when measured in accordance with ASTM D6489, both before and after artificial weathering to 4000 hours in accordance with ASTM G152/G153.

The paint system shall be resistant to the ingress of salt spray when measured in accordance with ASTM B117 after 4000 hours artificial weathering to ASTM G152/G153.

8.6 Concrete Subjected to Cyclic Immersion in Seawater

The coating applied to structures in the tidal zone may be subjected to abrasion from floating objects and possible contamination by oil and solvents.

The paint system shall be formulated for application onto dry but saturated concrete surface.

The paint system shall cure rapidly allowing for application and curing. As a practical minimum it is expected that the coating may be suitable for immersion in seawater within 3 hours of application.

The paint system shall be resistant to the diffusion of chloride ion when tested in accordance with AASHTO T259/ASTM C1202.

8.7 Concrete Subjected to Chemical Attack

Coatings may be immersed in strong acids, alkalis or solvents, subjected to a variety of chemicals, and may be subject to other reactive bacteria and cultures found in sewage.

Coating systems such as high performance solvent free epoxy and coal tar epoxy with at least 400 µm DFT are likely to be suitable.

Where the concrete structure is expected to crack, due to shrinkage or flexural movement, the paint system should be specified to be reinforced with glass-fiber fabric, of density 280-300 g/m^3 open weave lapped by 25 mm at its edges. In such circumstances, the minimum DFT of the system would be expected to exceed 1 mm.

The paint system shall be applied to at least two coats. The minimum DFT shall not be less than 275 µm and the target mean DFT shall be not less than 400 µm. This coating shall be tested in accordance with ASTM C267. The coating shall not deteriorate over a three-month test duration. The bond strength of the paint system measured in accordance with ASTM D7234, shall be satisfactory.

The water permeability of the paint system, measured in accordance with ASTM D870 shall be classified as resistant at the specified thickness.

8.8 Concrete Water Retaining Structures and Concrete subjected to Chloride Back-Penetration

The paint system shall comply with Section 8.4 and 8.9.

The paint system shall be resistant to the ingress of salt spray when measured in accordance with ASTM B117, after 4000 hours of artificial weathering.

8.9 Concrete Subjected to Active Cracks

The coating must have a substantial thickness, typically with a minimum DFT of at least 300 µm, and be elastomeric or plastomeric in physical terms in order to bridge cracks.

Elastomeric coatings for concrete are typically polyurethane based, whereas plastomeric are soft acrylic copolymer formulations, which commonly have high rates of dirt pick-up and may need cosmetic topcoats.

Coating system with crack bridging ability shall be preferred for concrete.

Crack-bridging paint systems shall not be applied over existing brittle paint finishes. Existing paint systems are to be removed prior to the application of crack resistant paint system whenever required. The choice of system to be used shall be subject to Employer approval.

The paint system shall comply with Section 7.3 and 8.4. Paints for application onto concrete with active cracks shall comply with the following:

i. The paint system shall not fail under 1000 cycles of flexure, when tested in accordance with ASTM C1305/C1305M at temperature of 5°C and 27°C at the minimum DFT.
ii. The paint system shall not fail in accordance with the requirement above after being subjected to artificial weathering for 4000 hours in accordance with ASTM G152/G153, at minimum DFT.

9. MANAGEMENT OF HAZARDOUS MATERIALS

9.1 General

The Contractor shall be responsible for the health and safety of all persons on the Contractor's worksite. All personnel in contact with the material shall be conversant with all safety procedures associated with the materials.

9.2 Material Safety Data Sheet

Material Safety Data Sheets (MSDS) and Product Data Sheets are to be kept on the work site for the duration of the work.

A copy of the MSDS must accompany all chemicals during transport.

All safety procedures recommended by the Manufacturer shall be adhered to particularly in relation to storage and handling of the coating system.

9.3 Manufacturer's Warning

All caution notices on the product containers, material labels and MSDSs shall be strictly observed and every necessary precaution taken to prevent accidents.

9.4 Statutory Authorities

RCER and any other relevant approvals must be obtained by the Contractor.

9.5 Thinners And Solvents

All thinners, solvents, primers and coating materials shall be regarded as hazardous materials and shall be used and stored properly according to the coating supplier's recommendations and Dangerous Goods Regulations.

9.6 Transport And Storage

Materials shall be transported, handled and stored in accordance with the Manufacturer's recommendations.

The materials shall be supplied, transported and stored in suitable containers or bags, which shall be clearly labeled. The labels shall not be removed or damaged in any way.

Unmarked containers or containers with broken seals shall not be allowed onto site and shall not be permitted to remain on site. Any damaged containers or containers without a suitable label shall be disposed of properly.

A store shall be provided on-site for the storage of hazardous materials. The store shall be covered, be enclosed by walls or a fence and shall be locked when not attended.

MSDSs shall be kept in the store. Fire extinguishers shall be kept adjacent to the store.

10. QUALITY ASSURANCE AND QUALITY CONTROL

10.1 General

For Quality requirements, refer to Ma'aden Capital Investment System (MCIS).

Unless approved in writing beforehand by the employer, all non-destructive testing (NDT) of welding shall be carried out by an independent authorized Testing Agency/Laboratory. All NDT operators shall be experienced and shall be trained and certified.

An Inspection and Test Plan (ITP) shall be prepared by the Contractor and submitted to the Employer for Approval prior to commencement of the works.

The ITP shall be part of the Project Quality Plan (PQP) which shall be submitted to the Employer for Approval prior to commencement of the works.

All work performed may be subject to inspection by the Employer or nominated representative.

The Contractor shall perform necessary inspection to assure that his daily work is meeting all requirements of this Specification.

The Contractor shall provide at least three days' notice prior to commencement of any work.

The Contractor shall complete the applicable Inspection Test Records (ITRs).

Prior to start of work, the Coating Manufacturer shall submit detailed written procedures for surface preparation and coating application for Employer's review and acceptance. Procedures shall demonstrate strict compliance with this specification and the latest Coating Manufacturer's product datasheet. Any inconsistency in this specification and Coating Manufacturer datasheet shall be submitted to Employer for approval before application of any coating.

Employer at its discretion may request Contractor to provide Coating Manufacturer representative at site to ensure coating application work is completed per Manufacturer recommendations.

Employer may also request Contractor at its discretion to perform additional tests at its own cost (not mentioned in this specification) in case deficiencies are observed in coating work performed by Contractor.

Necessary mockups of coating shall be carried out at site.

10.2 Quality Assurance and Traceability

All work and materials supplied shall be carried out to the requirements of ISO 9001 or Approved equivalent. Preference will be given to Coating suppliers and Contractors accredited to ISO 9001 or approved equivalent.

10.3 Hold Points

Hold Points shall be nominated in ITPs which shall be provided by the Contractor. Where a Hold Point is nominated, approval to proceed is required from the Employer.

The Hold Points shall include the following as a minimum requirement:

- Substrate preparation prior to the application of the primer
- Surface prior to the application of each resin layer
- Joint preparation prior to sealing.

10.4 Witness Points

Witness Points shall be nominated in ITPs which shall be provided by the Contractor. Where a Witness Point is nominated, the Employer shall be provided with adequate notice and opportunity to witness the work. The Witness Points shall include the following as a minimum requirement:

- Concrete soundness testing
- Joint rebate surface preparation
- Adhesion and thickness tests.

10.5 Contractor Records

The Contractor shall maintain written records of his work so that complete traceability of all work and materials provided under this Specification is maintained. Contractor shall ensure that the location of tests shall be well documented so that each test can be located easily in future.

These reports shall be maintained on a daily basis and submitted to the Employer when requested, or if not specifically requested at the completion of the project.

All testing records shall be attached to the ITP and shall form part of the Manufacturers Data Report (MDR).

10.6 Coating Application Record

The Contractor will be required to provide detail of coating application of the work site and include:

- The depth of concrete removal (measured from the original datum line)
- Locations at which the substrate was tested for pH and soundness. These locations are to be numbered to cross reference to the QC test report.
- The date of application and the batch number of each coat of material applied.

10.7 Film Thickness

The film thickness is the minimum dry film thickness, including primer coats specified in the coating system. Final acceptance of each increment of work will not be made until the dry film thickness meets or exceeds the specified thickness. Regardless of the number of coats specified, such

additional coats shall be applied as may be necessary to achieve the specified Total Coating Thickness (TCT).

The Contractor shall provide and operate wet film thickness gauges of an approved type to ensure the correct thickness of each coat, and the full system is achieved.

Final inspection of coated surfaces, including measurement of dry film thickness, may be performed or witnessed by the Employer or nominated representative.

For each area the Contractor shall demonstrate to the Employer, with the use of Quality Assurance/Quality Control records, including comparison of actual and theoretical application rates and wet film thickness measurements, that the nominated dry film thickness has been achieved.

10.8 Correction of Defects and Deficiencies

All deficiencies and defects revealed by the above inspection and test methods shall be promptly repaired or corrected, as per Coating Manufacturer recommendations and as approved by Employer.

10.9 Inspector

A certified inspector of coatings shall be appointed for inspection and testing of protective coating systems by the Contractor.

The inspector is to be nominated by the Contractor and approved by the Employer prior to the commencement of work.

All work may be subject to inspection by the Employer or his nominated representative; however, this shall not relieve the Contractor of their own internal Quality Assurance/Quality Control responsibilities.

10.10 Adhesion Test of Completed Coating System

Adhesion testing shall be carried out on the completed coating system. Test using an Adhesion Pull Off Tester as described in ASTM D4541. The test dolly must show a 90% concrete residue at the delaminate interface and an adhesion strength of not less than the greater of:

- 1.0 MPa
- The adhesion strength nominated as being required by the Coating Supplier.

Each test point is to be repaired immediately using the full protective coating system.

10.10.1 *Frequency of Testing*

An adhesion test of the completed coating system shall be carried out at least once in each 40 m2 area or at least once on each discrete section. A discrete section includes isolated areas or areas bound by external perimeter and/or expansion or control joints.

An adhesion test of the completed coating system shall be carried out at least once in each 20 lineal meters of vertical surfaces of bund walls, plinths, columns, pump bases, and other up stands.

Holiday testing shall be conducted in accordance with NACE SP0188. For immersion services, 100% of coated area shall be inspected. Cross cut test shall be carried out as per ISO 2409.

Additional tests shall be carried out at the discretion of the Employer.

10.10.2 *Continuity Testing Using a High Voltage Holiday Detection Device*

Where nominated in the scope of work, on contract drawings or elsewhere in the contract documents, all coated surfaces shall be tested using a high voltage holiday detection device.

The test voltage shall be selected having regard to the specified minimum dry film thickness and the dielectric properties of the coating.

Defects in the coating and nature of the defect shall be marked on a plan of the area.

Defective areas shall be re-tested after completion of repairs.

10.10.3 *Quality Management Program*

For the Quality Management Program, the following submissions shall be made as a prequalification package for the coating system:

- Approved 3rd party lab test results and certificates of compliance from Manufacturer
- Coding System
- Storage Life
- Method of using paint/coating component
- Surface preparation of concrete requirement for particular type of coating
- Dry film thickness and coverage
- Over Coating Time
- Physical properties data sheet
- Durability data
- Health and safety datasheet.

Appendix A

Approved Protective Coating Systems for Concrete & Concrete Masonry

Table A-1: Coating Systems for Concrete Masonry

Item	S.No.	Generic Coating System	Min. Dry Fil thickness (Microns)
Coatings on Exterior Surface of Concrete Masonry	1	Primer: Waterborne Cementitious Acrylic	Applied to fill all voids
		Intermediate: HDP Acrylic Polymer	50 - 75
		Finish Coat: HDP Acrylic Polymer	50 - 75
	2	Primer: Acrylic Emulsion	200 - 250
		Second Coat: Acrylic Emulsion	200 - 250
	3	Primer: Waterborne Cementitious Acrylic	Applied to fill all voids
		Intermediate: Polyamidoamine Epoxy	100 - 150
		Finish Coat: Waterborne Acrylic Polyurethane	50 - 75
	4	Primer: Waterborne Cementitious Acrylic	Applied to fill all voids
		One Coat: Acrylic Emulsion	Minimum 200 - 250
Coatings on Interior Surface of Concrete Masonry	1	Primer: Waterborne Cementitious Acrylic	Applied to fill all voids.
		Intermediate: HDP Acrylic Polymer	50 - 75
		Finish Coat: HDP Acrylic Polymer	50 - 75
	2	Primer: Waterborne Cementitious Acrylic	Applied to fill all voids
		Finish Coat: Polyamidoamine Epoxy	100 - 150
	3	Primer: Waterborne Cementitious Acrylic	Applied to fill all voids
		Intermediate: Polyamidoamine Epoxy	100 - 150
		Finish Coat: Waterborne Acrylic Polyurethane	50 - 75

Table A-2: Coating Systems for Concrete Structures

System	Item	Generic Coating System	Min. Dry film Thickness (Microns)	Information on Area to be coated and Other Remarks
1	Above ground concrete structures in plants	1st Coat: Sealer coat	N/A	Coating should be pigmented with Aluminium flakes. Grey colour is to be used.
		2nd Coat: One pack moisture cure Polyurethane coating	500	For extreme service conditions such as: acidic environment,
		3rd Coat: One pack moisture cure Polyurethane coating	500	For Alkaline environment, spillage, or wet conditions use system#3
2	Below ground concrete structures in plants	1st Coat: Sealer coat	N/A	Apply according to paint manufacturer's instructions and specification for concrete substrate.
		2nd Coat: Solvent free Amine cured Epoxy	500	
3	Concrete chambers containing raw seawater/brackish water, neutralization pond or concrete	1st Coat: Sealer coat	N/A	Apply according to paint manufacturer's instructions and specification
		2nd Coat: Two pack solvent free Polyurethane coating	600	

System	Item	Generic Coating System	Min. Dry film Thickness (Microns)	Information on Area to be coated and Other Remarks
	foundations in low water table areas	3rd Coat: Two pack solvent free Polyurethane coating	600	
4	Concrete in seawater services (e.g. intake and outtake chambers)	1st Coat: Primer / sealer Amine cured Epoxy	N/A	Apply according to paint manufacturer's instructions and specification
		2nd Coat: Solvent free Amine cured Epoxy[1]	2 x 500	For better flexibility and abrasion resistance paint system#3 can be used.
5	Concrete in seawater services (with abrasive particles)	1st Coat: SP/ HS Glass flake Epoxy coating	500	
		2nd Coat: SP/ HS Glass flake Epoxy coating	500	
6	Concrete floor in warehouse, workshop, paving etc. in plants.	1st coat: Epoxy Polyamide/Amine cured sealer/primer	N/A	Apply according to paint manufacturer's instructions and specification
		2nd coat: High build Polyamide Epoxy[2]	150	
		3rd coat: High build Polyamide Epoxy	125	
Notes	1. Applied in two coats or in one coat 2. For non-slip surface scatter antislip aggregate into the wet coat. Graded, washed and dust free Silica sand or Garnet is suitable.			